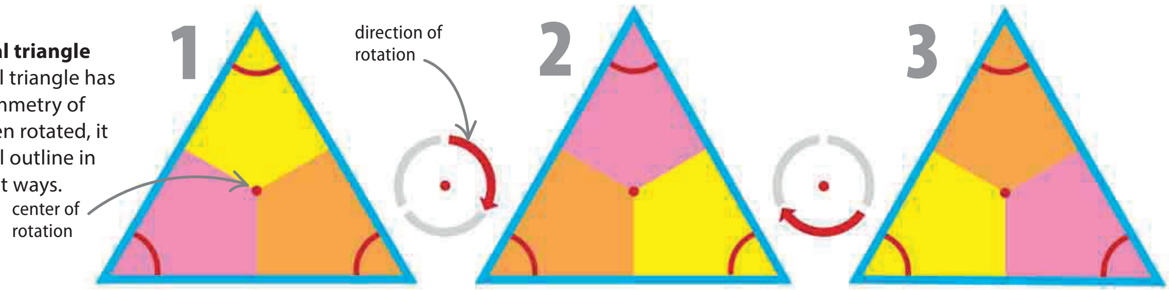
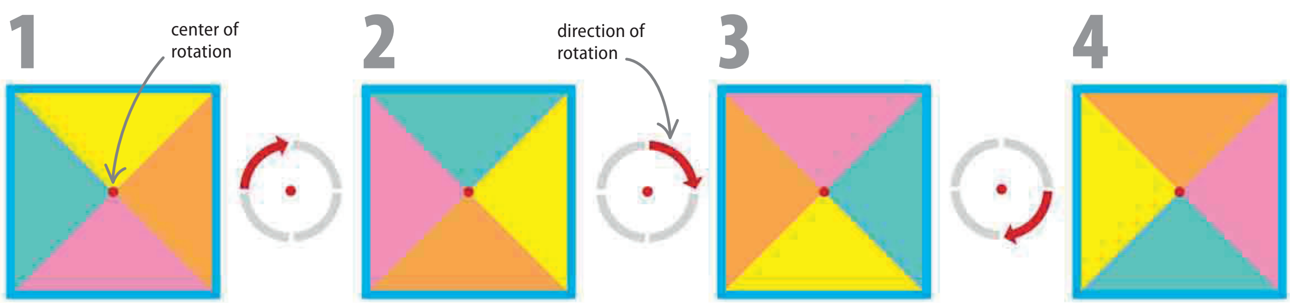
Rotational symmetry

A two-dimensional shape has rotational symmetry if it can be rotated about a point, called the center of rotation, and still exactly fit its original outline. The number of ways it fits its outline when rotated is known as its "order" of rotational symmetry.

▷ **Equilateral triangle**
An equilateral triangle has rotational symmetry of order 3—when rotated, it fits its original outline in three different ways.



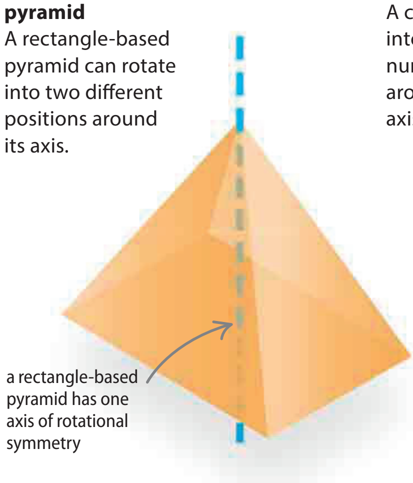
▽ **Square**
A square has rotational symmetry of order 4—when rotated around its center of rotation, it fits its original outline in four different ways.



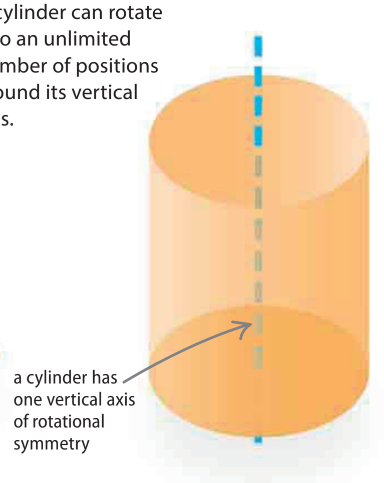
Axes of symmetry

Instead of a single point as the center of rotation, a three-dimensional shape is rotated around a line known as its axis of symmetry. It has rotational symmetry if, when rotated, it fits into its original outline.

▽ **Rectangle-based pyramid**
A rectangle-based pyramid can rotate into two different positions around its axis.



▽ **Cylinder**
A cylinder can rotate into an unlimited number of positions around its vertical axis.



▽ **Cuboid**
A cuboid can rotate into two different positions around each of its three axes.